

Lab2

Interconnexion of two LANS with two routers

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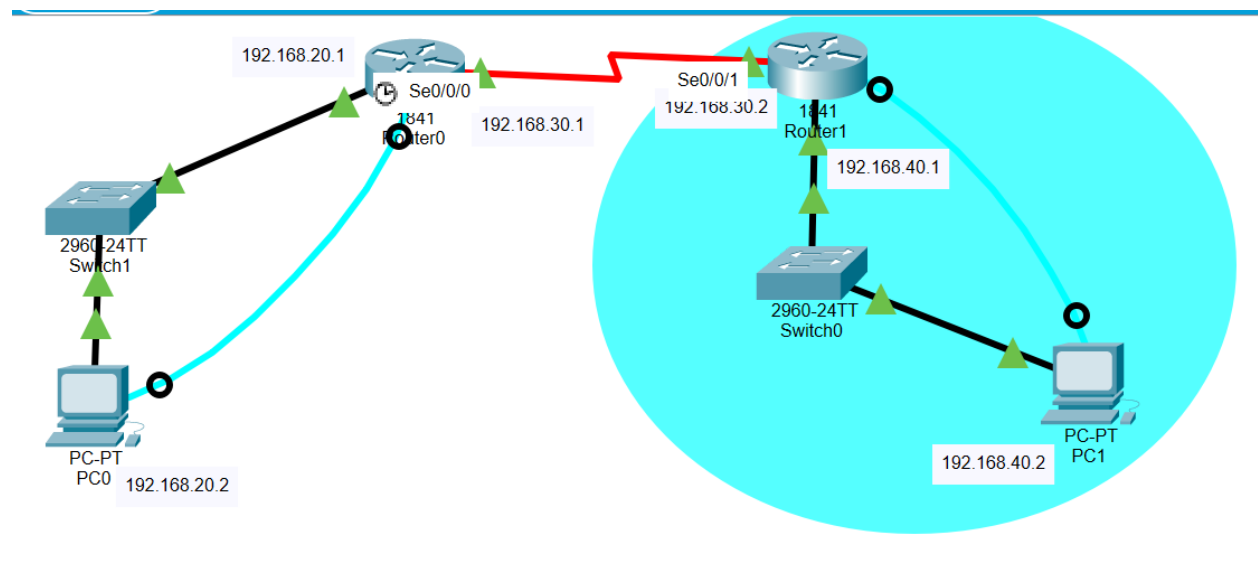
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OBJECTIVES:

1. To understand why routers must know all networks in a topology to forward packets correctly.
2. To explain how devices in different LANs communicate through a router and how routers are connected over a WAN using serial links.
3. To configure routing using static routes and RIP, and compare how each method shares network information.
4. To implement and use a firewall to control traffic and improve network security.

TOPOLOGY:



Materials used:

1. Devices

- *Router* – Used to connect different networks and route traffic between them.
- *Switch* – Used to connect devices within the local area network (LAN).
- *PC* – Used as an end device for configuration, testing, and communication.

2. Cables

The following cables were used to connect the devices:

- *RJ45 Ethernet Cable* – Used to connect the PC to the switch and the switch to the router.
- *Console Cable* – Used to connect the PC to the router for configuration and management.
- *Serial Cable* – Used to connect one router to another router for WAN communication.

CABLING:

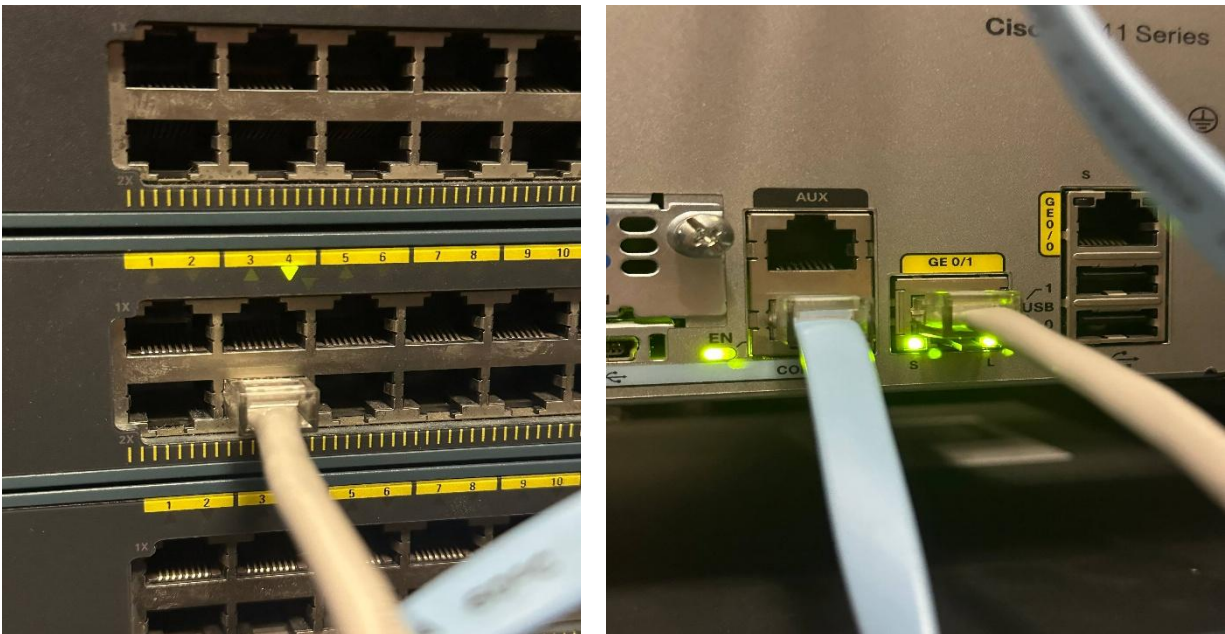
1-Router to Router:

I connected the serial cable to the **SO/0/1** interface on my router.



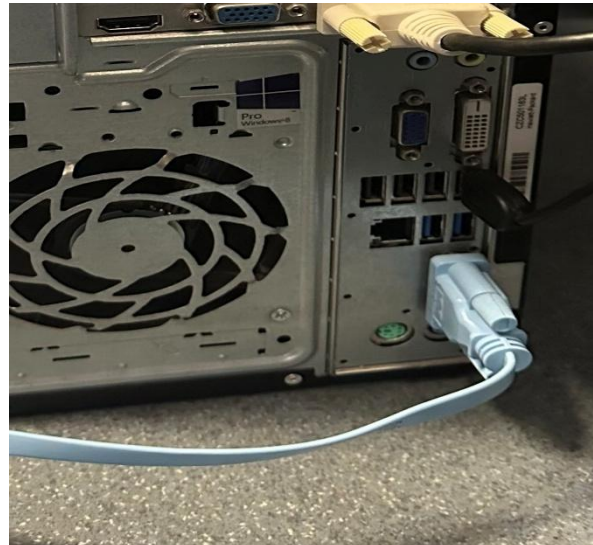
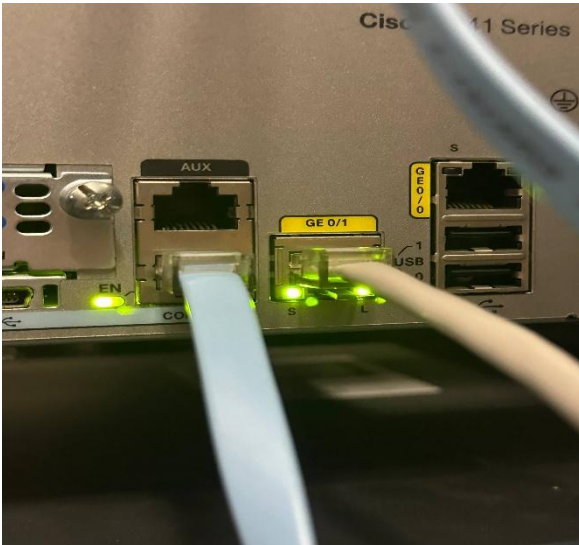
2-Router to Switch:

I connected the switch to the router on the **G0/1** interface.



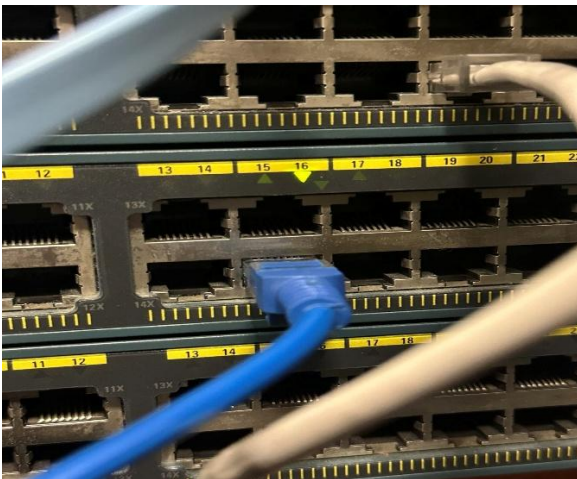
3-Router to PC:

I connected the Router to PC .



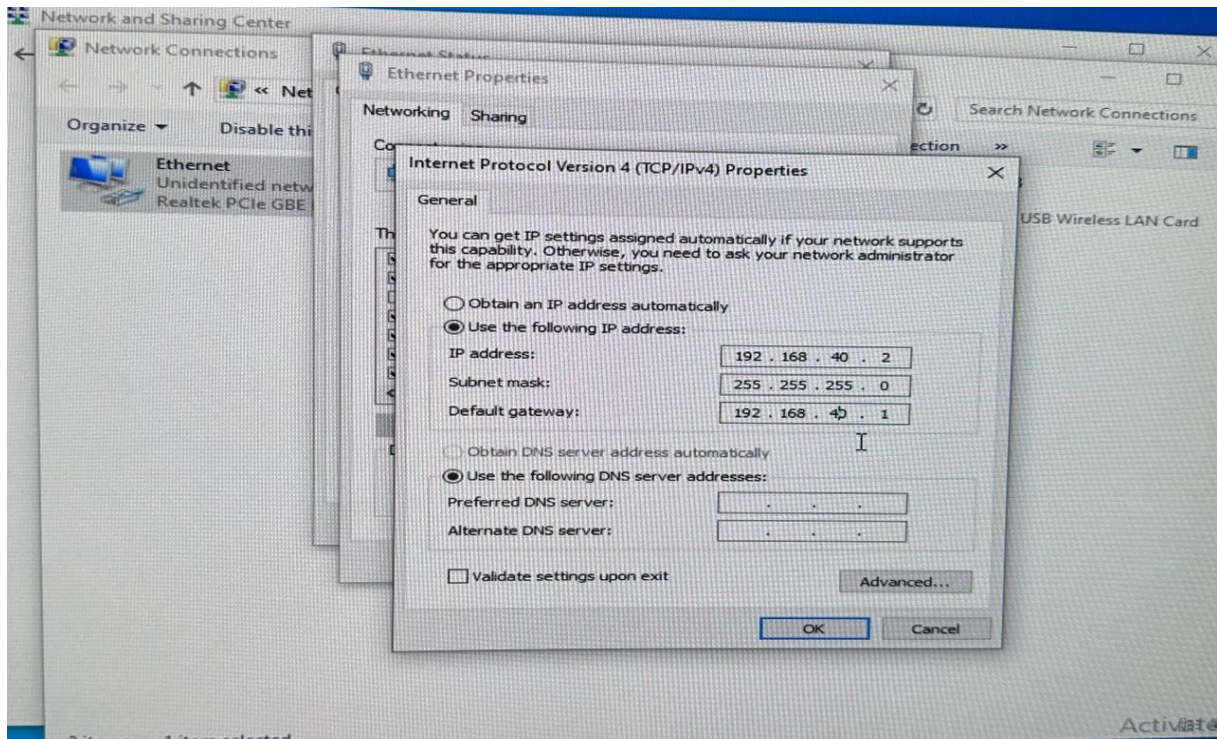
4-Switch to PC:

I connected the Switch to PC.



ASSIGN IP ADDRESS TO PC:

Open Control Panel → Go to Network and Sharing Center → Click Change adapter settings → Right-click Ethernet → Select Properties → Choose Internet Protocol Version 4 (TCP/IPv4) → Click Properties

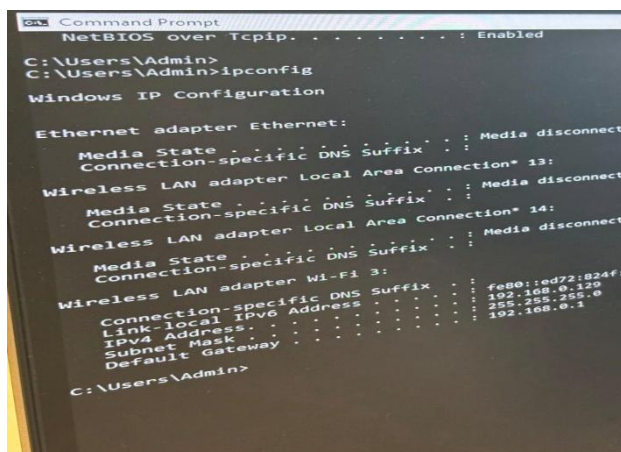


I open Command Prompt to check ip of the pc:

I write the command **ipconfig**

The problem:

IP of my pc NOT SHOWING Because the Ethernet cable was not connected



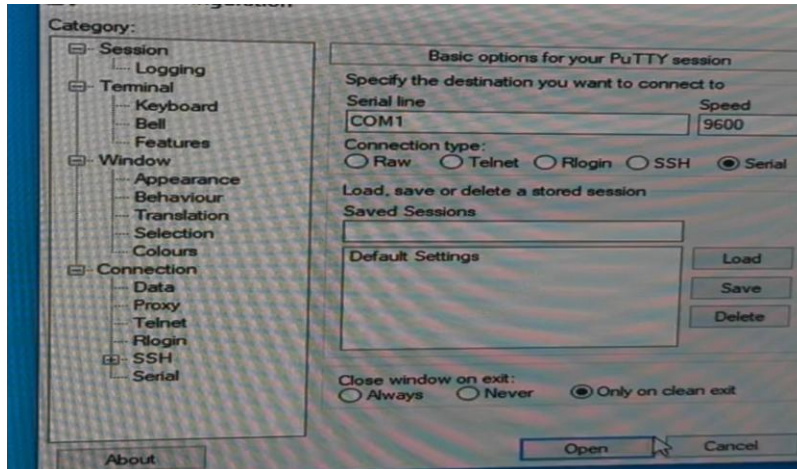
The solution :

I connected the cable After that the **ip** appear



Router Configuration:

Using PuTTY, we connected to the router and typed commands to set it up.



Entering the username: cisco and password: cisco for security purposes in putty.

```
R2(config)#  
R2(config)#username cisco  
R2(config)#enable secret cisco  
R2(config)#  
R2(config)#
```

Check if the serial are DTE or DCE:

Write the command **show controllers serial 0/0/1**, and I see the serial is **DTE**.

```
R2#show controllers serial 0/0/1  
Interface Serial0/0/1  
Hardware is SCC  
DTE V.35  
Clock Freq detected Rx clk/Tx clk 193  
  
ldb at 0x2B9169EC, driver data struct  
wic_info 0x2B9178FC  
  
WIC Serial Global Registers: (0x10490  
-----  
sti_errintr_en      (0004): 0x00000000  
mgmt_intr_status    (1010): 0x0010  
mgmt_intr_enable     (1020): 0x00CA  
timer_0_7_enable     (1024): 0x0000  
timer_0_1_program    (1030): 0x0000
```

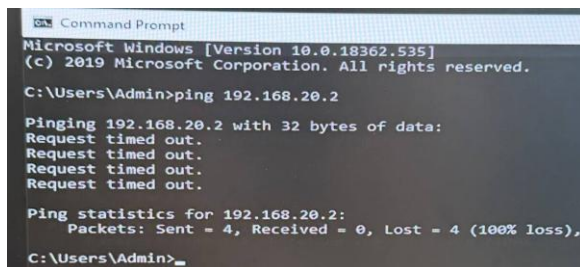
Configure the serial interface s0/0/1:

```
R2(config)#interface s0/0/1
R2(config-if)#no shutdown
R2(config-if)#ip address 192.168.30.2 255.255.255.0
R2(config-if)#
```

Configure interface g0/1:

```
R2(config)#
R2(config)#interface g0/1
R2(config-if)#ip address 192.168.40.1 255.255.255.0
R2(config-if)#
```

Ping to the another pc in different LAN:



```
Command Prompt
Microsoft Windows [Version 10.0.18362.535]
(c) 2019 Microsoft Corporation. All rights reserved.

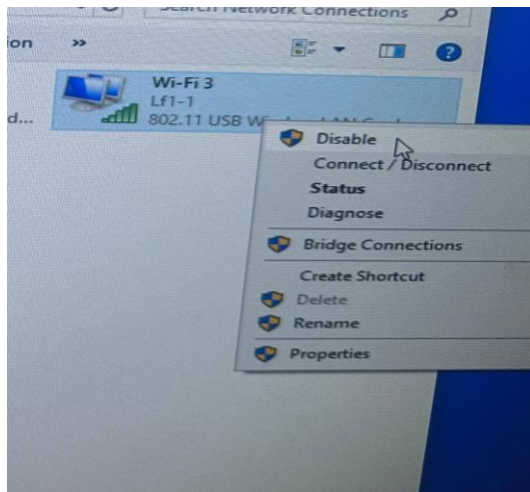
C:\Users\Admin>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\Users\Admin>
```

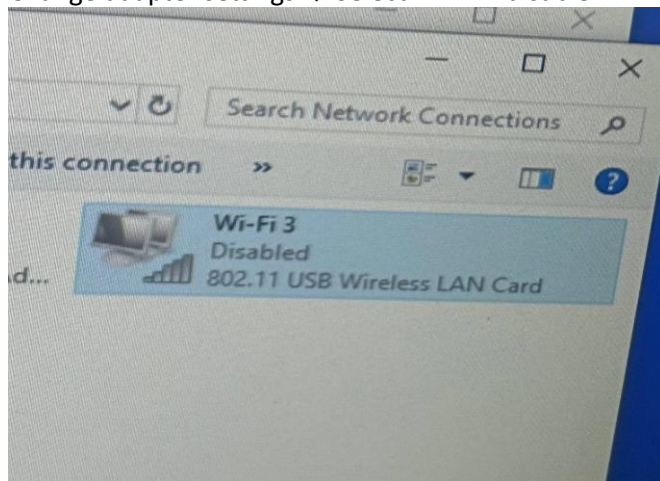
I think this is a problem so I try to ping to my router :

But the **BIG PROBLEM** is the ping is also not working because the wifi is enable:



The **SOLUTION** is :

Control Panel → Network and Sharing Center →
Change adapter settings → Select Wi-Fi ->disable.



Configure Static Route :

This command manually tells R1 to forward traffic destined to 192.168.20.0/24 through the next-hop IP address 192.168.30.1

```
R2(config)#  
R2(config)#ip route 192.168.20.0 255.255.255.0 192.168.30.1  
R2(config)#  
R2(config)#
```

NOW I TRY TO PING THE ANOTHER PC:

```
C:\Users\Admin>ping 192.168.20.2  
  
Pinging 192.168.40.2 with 32 bytes of data:  
Reply from 192.168.40.2: bytes=32 time<1ms TTL=128  
Reply from 192.168.40.2: bytes=32 time<1ms TTL=128  
Reply from 192.168.40.2: bytes=32 time<1ms TTL=128  
Reply from 192.168.40.2: bytes=32 time<1ms TTL=128
```

The ping worked.

DYNAMIC ROUTE(RIP):

I remove the static route using the command

No ip route 192.168.20.0 255.255.255.0 192.168.30.1

And write the commands to configure RIP:

```
R2(config)#router rip  
R2(config-router)#version 2  
R2(config-router)#network 192.168.40.0  
R2(config-router)#network 192.168.30.0  
R2(config-router)#RIP: sending v2 request to 224.0.0.9 via Serial0/0/1 (192.168.30.2)  
RIP: sending v2 update to 224.0.0.9 via GigabitEthernet0/1 (192.168.40.1)  
RIP: build update entries  
192.168.30.0/24 via 0.0.0.0, metric 1, tag 0  
RIP: sending v2 update to 224.0.0.9 via Serial0/0/1 (192.168.30.2)  
RIP: build update entries  
192.168.40.0/24 via 0.0.0.0, metric 1, tag 0
```

AND I try ping -> ping worked

```
C:\Users\Admin>ping 192.168.20.2  
  
Pinging 192.168.40.2 with 32 bytes of data:  
Reply from 192.168.40.2: bytes=32 time<1ms TTL=128  
Reply from 192.168.40.2: bytes=32 time<1ms TTL=128  
Reply from 192.168.40.2: bytes=32 time<1ms TTL=128  
Reply from 192.168.40.2: bytes=32 time<1ms TTL=128
```


FIREWALL:

The second teams put firewall and I try to ping them but didn't worked.

```
C:\Users\Admin>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

But the other team tried to ping my PC, and it worked. So, I enabled the firewall on my PC to block the ping requests.

Open Control Panel → Click System and Security → Click Windows Defender Firewall → Click Turn Windows Defender Firewall on or off

Customize settings for each type of network

You can modify the firewall settings for each type of network that you use.

Private network settings

- ☒ Turn on Windows Defender Firewall
 - ☐ Block all incoming connections, including those in the list of allowed apps
 - ☒ Notify me when Windows Defender Firewall blocks a new app

- ☐ Turn off Windows Defender Firewall (not recommended)

Public network settings

- ☒ Turn on Windows Defender Firewall
 - ☐ Block all incoming connections, including those in the list of allowed apps
 - ☒ Notify me when Windows Defender Firewall blocks a new app

- ☐ Turn off Windows Defender Firewall (not recommended)

Select Turn on Windows Defender Firewall → Click OK

Customize settings for each type of network

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Public network settings

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 - ☐ Block all incoming connections, including those in the list of allowed apps
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- ☐ Turn off Windows Defender Firewall (not recommended)

Finally, no one was able to ping the other devices after the firewall was enabled.

CONCLUSION:

In this lab, we learned how routers connect different networks and allow devices to communicate. We configured IP addresses and used PuTTY to set up the routers.

We configured both static routing and dynamic routing (RIP) to connect the networks. Static routing was set manually, while dynamic routing shared information automatically.

At first, devices could ping each other. After we enabled the firewall, ping was blocked. In the end, none of the devices could ping each other, which showed that the firewall was working correctly.